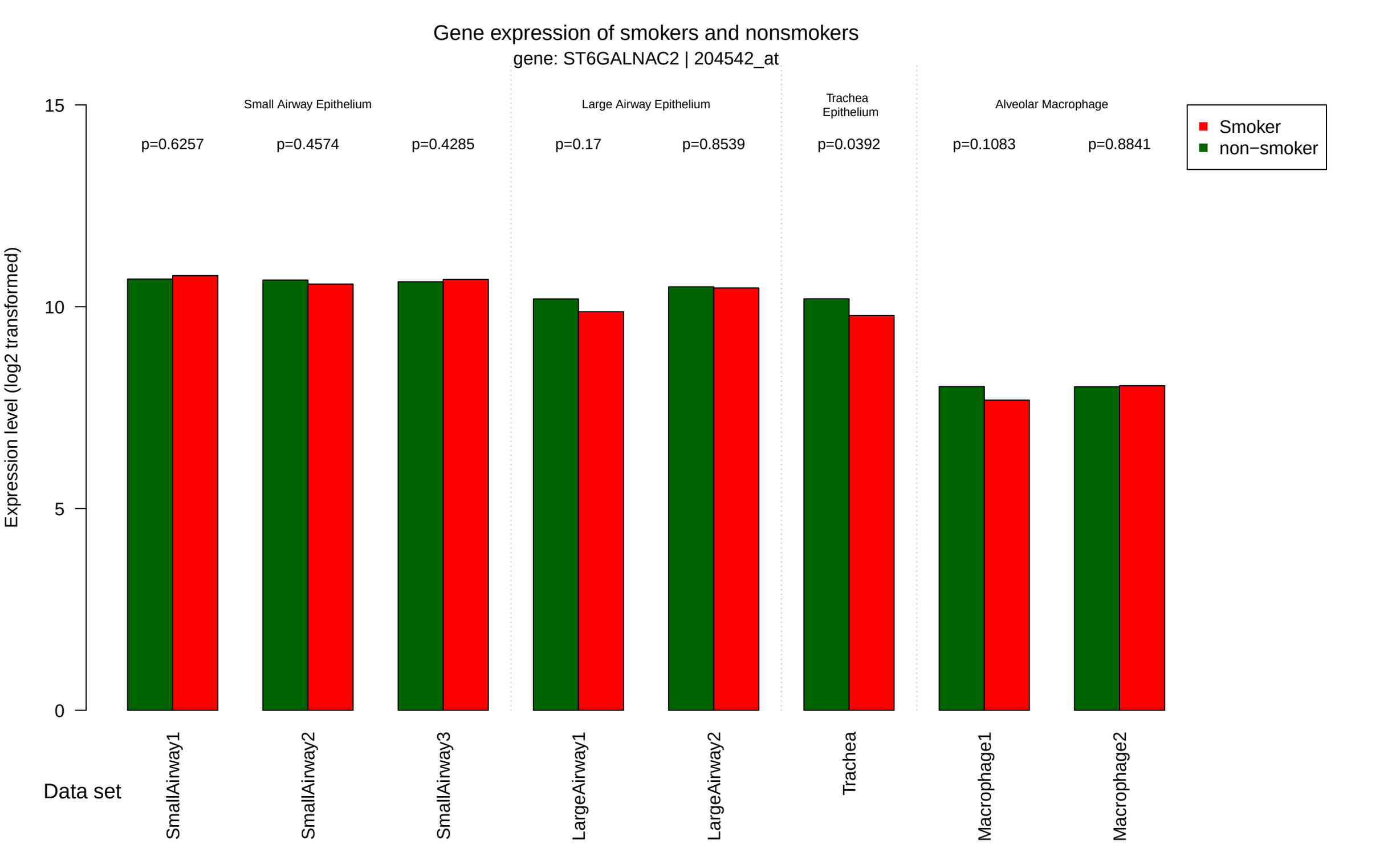
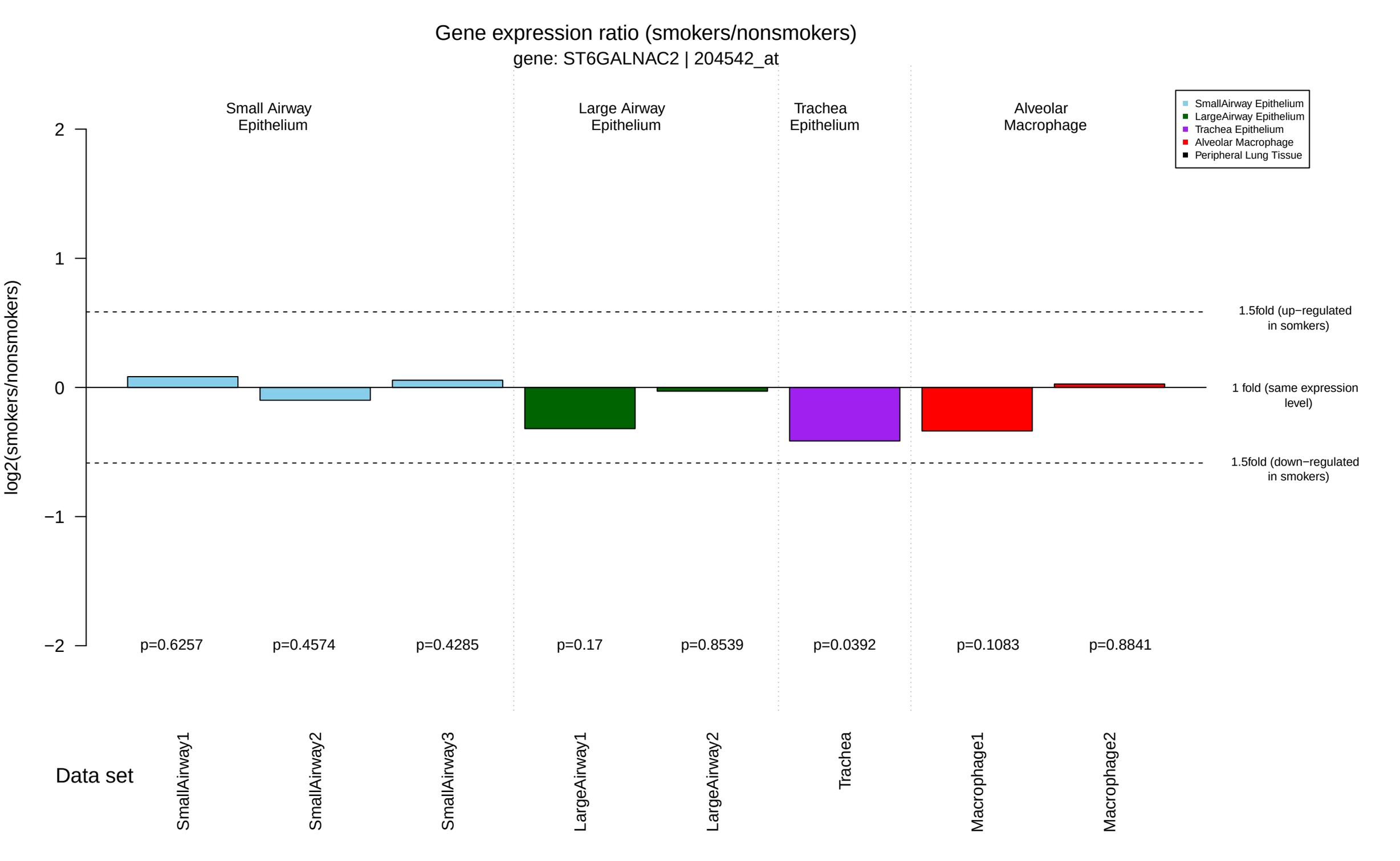
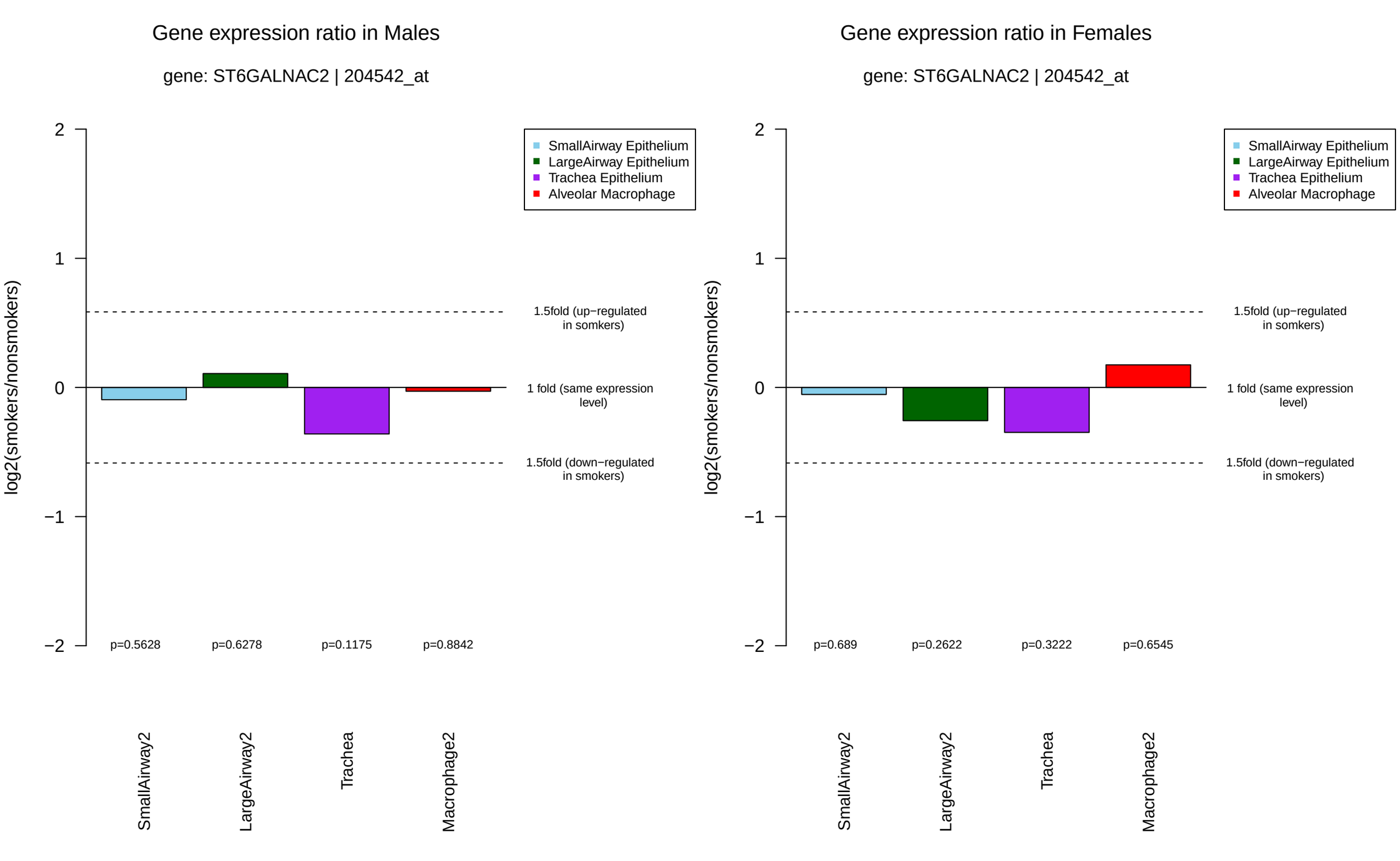
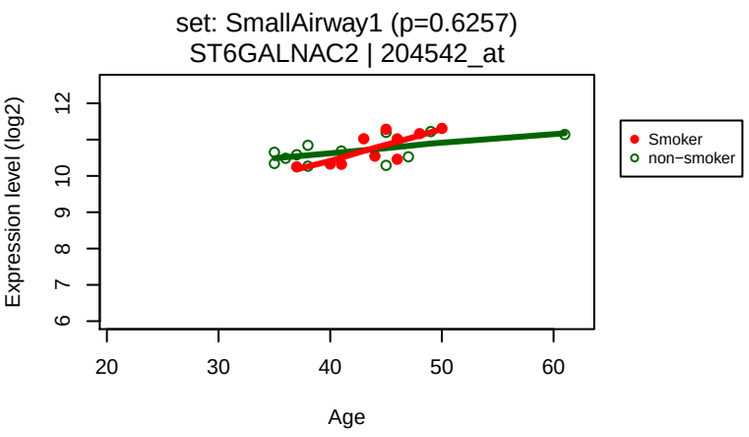
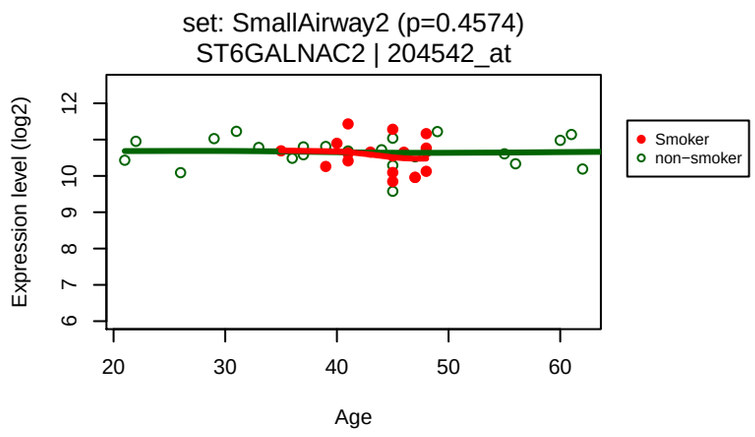
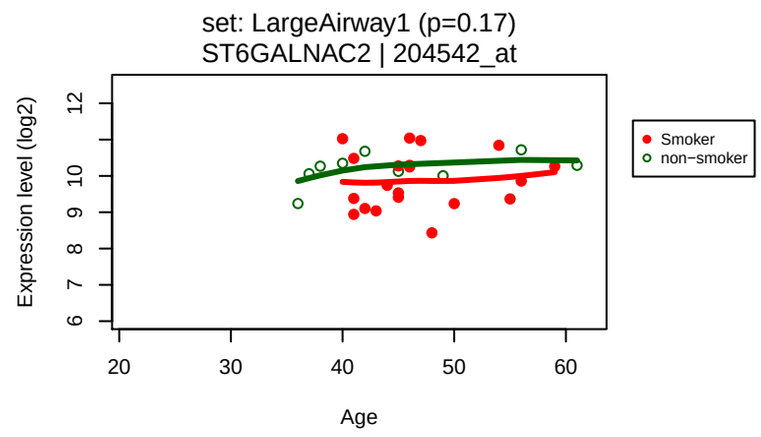

Gene expression of smokers and nonsmokers
gene: ST6GALNAC2 | 204542_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6257
p=0.4574
p=0.4285
p=0.17
p=0.8539
p=0.0392
p=0.1083
p=0.8841
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

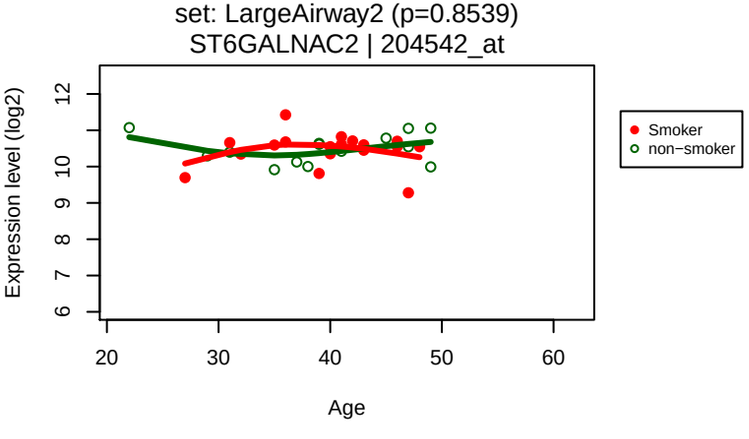

Gene expression ratio (smokers/nonsmokers)
gene: ST6GALNAC2 | 204542_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6257
p=0.4574
p=0.4285
p=0.17
p=0.8539
p=0.0392
p=0.1083
p=0.8841
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

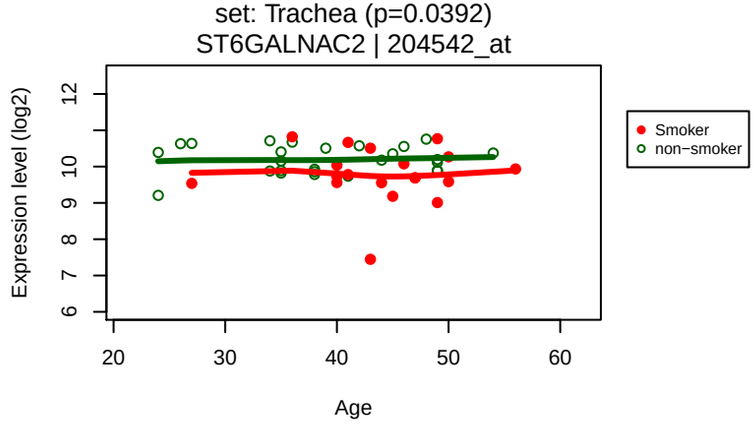

Gene expression ratio in Males
gene: ST6GALNAC2 | 204542_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5628
p=0.6278
p=0.1175
p=0.8842
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST6GALNAC2 | 204542_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.689
p=0.2622
p=0.3222
p=0.6545
SmallAirway2
LargeAirway2
Trachea
Macrophage2

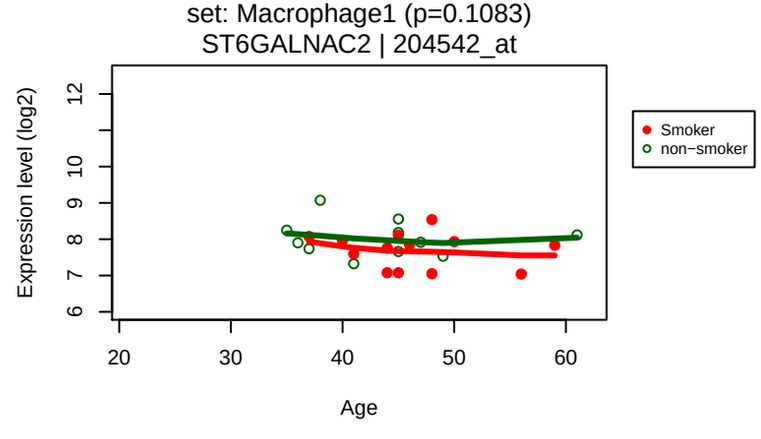

set: SmallAirway1 (p=0.6257)
ST6GALNAC2 | 204542_at
Expression level (log2)
Age
Smoker
non−smoker

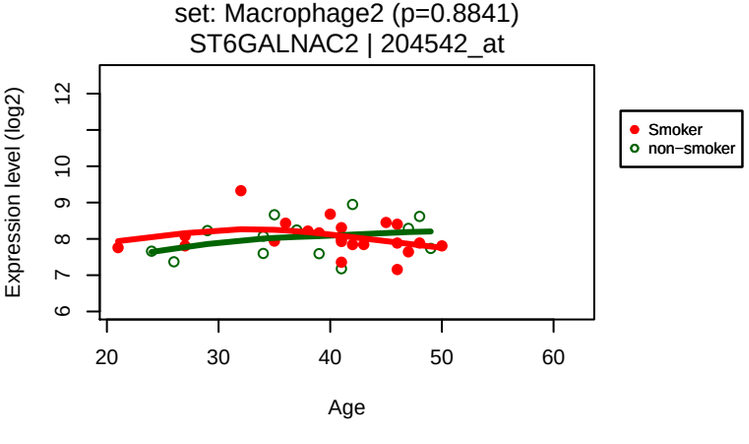

set: SmallAirway2 (p=0.4574)
ST6GALNAC2 | 204542_at
Expression level (log2)
Age
Smoker
non−smoker

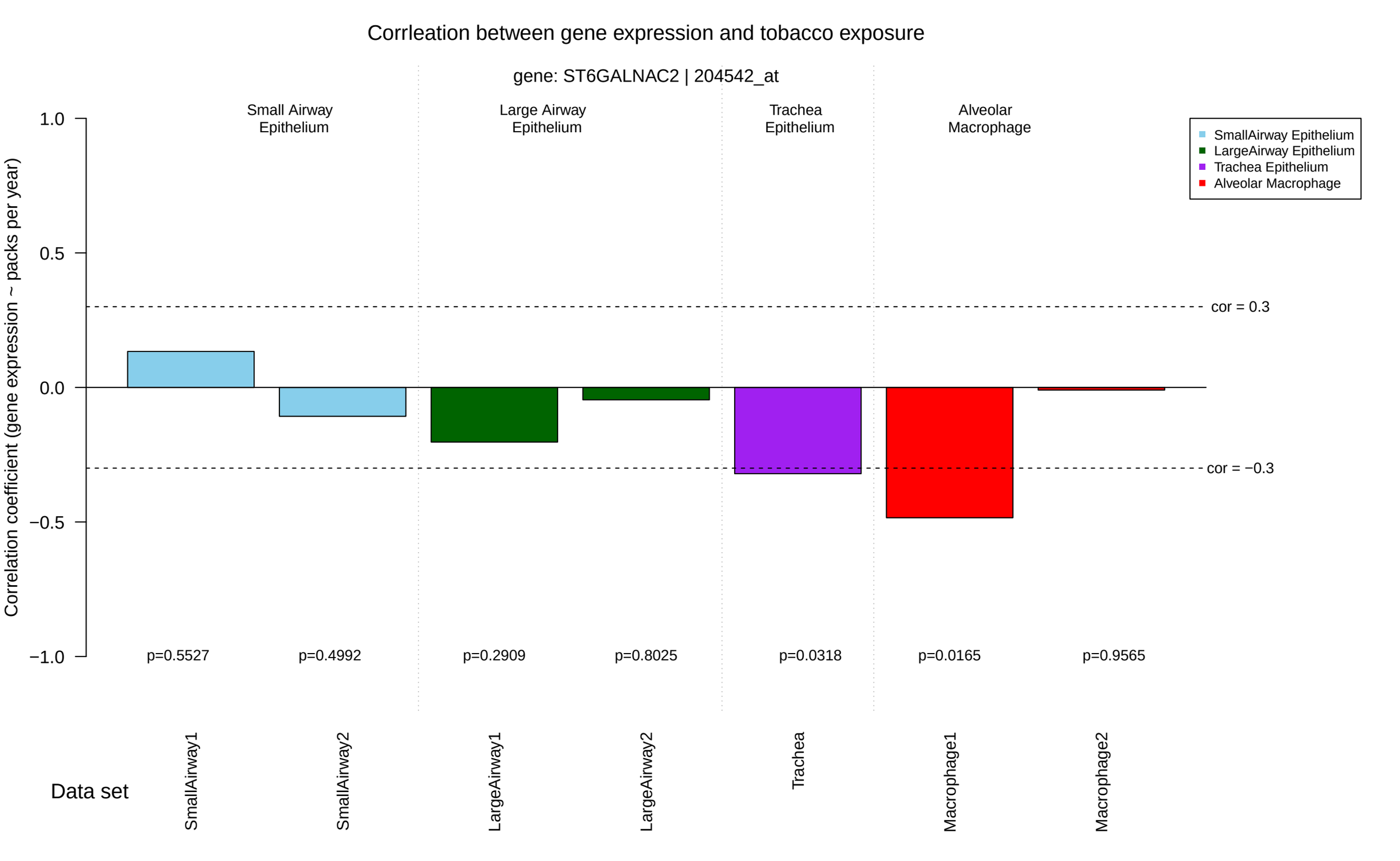

set: LargeAirway1 (p=0.17)
ST6GALNAC2 | 204542_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8539)
ST6GALNAC2 | 204542_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0392)
ST6GALNAC2 | 204542_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1083)
ST6GALNAC2 | 204542_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8841)
ST6GALNAC2 | 204542_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST6GALNAC2 | 204542_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5527
p=0.4992
p=0.2909
p=0.8025
p=0.0318
p=0.0165
p=0.9565
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2